

ROCHESTER INTL, MN, USA

WMO: 726440

Lat: 43.904N

Lon: 92.492W

Elev: 398

StdP: 96.64

Time Zone: -6.00 (NAC)

Period: 94-19

WBAN: 14925

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-24.9	-22.3	-28.3	0.3	-24.5	-25.8	0.4	-21.9	14.2	-10.6	12.7	-9.0	5.0	310	0.645

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.1	31.0	23.1	29.3	22.1	27.9	21.4	25.0	29.0	23.7	27.7	22.7	26.5	6.1	200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
23.6	19.4	27.8	22.4	17.9	26.5	21.3	16.8	25.3	78.1	29.0	73.0	27.7	68.9	26.4	29.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	(q)
(4)	12.5	11.2	10.1	DB	-28.5	33.7	3.2	1.6	-30.8	34.9	-32.6	35.9	-34.4	36.8	-36.8	38.0
(5)				WB	-28.5	26.6	2.9	1.2	-30.6	27.5	-32.3	28.2	-34.0	28.9	-36.1	29.7

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	7.3	-9.5	-7.4	-0.2	7.5	14.2	19.7	21.6	20.2	16.5	9.2	1.2	-6.2
	DBStd	12.21	7.23	6.91	6.88	5.54	4.82	3.65	3.04	2.95	4.43	5.31	6.07	6.57
	HDD10.0	2393	606	487	322	109	14	0	0	4	80	269	502	
	HDD18.3	4322	864	719	574	325	145	27	7	14	86	287	514	760
	CDD10.0	1416	0	0	7	35	146	292	358	316	200	56	6	0
	CDD18.3	303	0	0	1	1	18	69	107	72	32	4	0	0
	CDH23.3	2376	0	0	2	23	182	553	848	482	247	39	0	0
	CDH26.7	602	0	0	0	4	54	153	233	96	54	8	0	0
Wind	WSAvg	5.1	5.8	5.7	5.6	5.9	5.4	4.7	4.0	3.7	4.3	5.1	5.6	5.6
Precipitation	PrecAvg	803	22	21	48	79	97	113	104	102	88	58	42	29
	PrecMax	1116	64	56	101	185	311	318	313	358	267	192	150	93
	PrecMin	392	2	1	8	24	30	24	26	9	10	7	2	6
	PrecStd	162	14	14	24	39	50	60	58	57	64	37	32	18
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	6.2	9.6	21.4	26.5	31.5	32.7	33.0	31.2	30.7	27.7	19.4	11.3
		MCWB	3.6	7.2	16.1	16.8	19.9	23.1	25.3	24.7	22.2	18.9	13.3	9.7
	2%	DB	3.2	5.9	16.1	22.2	27.8	30.4	30.9	29.0	28.2	23.5	15.4	6.6
		MCWB	1.5	4.3	12.4	13.8	18.0	21.7	24.4	23.0	21.1	16.5	11.0	4.9
	5%	DB	1.4	3.4	12.5	19.3	25.0	28.3	29.1	27.6	26.2	20.5	12.4	3.8
		MCWB	0.4	1.9	9.3	12.1	17.1	20.6	23.0	22.4	19.7	15.1	9.0	2.4
	10%	DB	0.1	1.7	8.9	16.5	22.4	26.5	27.7	26.1	24.2	17.5	9.8	2.0
		MCWB	-0.6	0.5	6.4	10.9	15.7	19.7	22.1	21.2	18.5	13.0	6.8	0.7
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	4.1	7.7	16.7	18.0	22.2	25.0	27.2	25.8	24.0	20.7	14.5	10.2
		MCDB	6.1	9.3	20.2	24.5	29.0	30.4	31.0	29.4	28.4	24.2	18.0	11.5
	2%	WB	1.7	4.4	13.1	15.3	19.9	23.4	25.5	24.2	22.3	17.8	12.0	4.9
		MCDB	2.9	5.8	15.6	20.4	25.1	28.1	29.4	27.7	26.5	21.5	14.8	6.3
	5%	WB	0.5	2.0	9.6	13.2	18.4	22.1	24.2	23.1	20.9	15.7	9.2	2.5
		MCDB	1.3	3.2	12.5	18.0	23.0	26.7	28.2	26.4	24.8	19.2	12.2	3.7
	10%	WB	-0.6	0.6	6.4	11.3	16.8	20.9	23.0	22.1	19.6	13.6	7.0	0.8
		MCDB	0.2	1.6	8.8	15.7	21.0	25.2	26.8	25.2	23.0	17.4	9.6	1.8
Mean Daily Temperature Range	5% DB	MDBR	8.0	8.3	8.6	10.4	10.8	10.2	10.1	10.0	11.1	10.1	8.4	7.5
		MCDBR	8.4	8.3	12.5	15.1	14.1	12.3	11.1	10.8	12.8	13.8	12.3	8.8
		MCWBR	7.4	7.0	8.7	8.7	7.0	6.0	5.9	6.1	6.7	7.7	8.5	7.4
	5% WB	MCDBR	7.8	7.9	11.9	13.4	12.1	10.5	10.2	9.9	11.2	11.3	11.3	8.3
		MCWBR	7.0	6.8	8.6	8.6	6.9	6.2	6.1	6.1	6.7	7.6	8.4	7.2
Clear-Sky Solar Irradiance	taub	0.262	0.291	0.314	0.363	0.390	0.400	0.405	0.397	0.366	0.321	0.287	0.261	
	taud	2.415	2.325	2.395	2.314	2.292	2.301	2.287	2.320	2.382	2.494	2.506	2.450	
	Ebn at Noon	880	903	926	900	881	871	861	856	858	861	844	848	
	Edh at Noon	73	96	103	122	129	128	128	120	104	82	68	64	
All-Sky Solar Radiation	RadAvg	1.75	2.81	3.76	4.57	5.38	6.00	6.26	5.29	4.21	2.71	1.77	1.30	
	RadStd	0.19	0.30	0.37	0.47	0.41	0.46	0.26	0.35	0.30	0.37	0.16	0.12	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (51 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page